Atharv Patil

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Education

The Pennsylvania State University - UP — *Aug 2021 - May 2025*
B.S. in COMPUTER SCIENCE

Courses: Artificial Intelligence, Distributed Systems, Operating Systems, DBMS, DSA, OOP, Theory of Computation, Systems Programming

Work Experience

Amphenol Corp. — *Nashua, NH*
Software Engineer Intern — *Jun 2024 - Aug 2024*

- Developed a ***C++*** command-line interface to communicate with PWM and thermocouple amplifier I2C chips, enabling accurate, production-grade thermal testing for high-speed AWS server cables
- Designed modular wrapper functions based on chip datasheets, allowing full feature access, robust control, and future scalability
- Built test environment using PWM logic to emulate real testing conditions for empirical analysis prior to thermostream system integration
- Implemented a custom PID control algorithm, reducing temperature deviation from ±7°C to ±0.5°C and surpassing the customer's strict ±1°C accuracy requirement

Academic Excellence Center (AEC), Penn State — *State College, PA*
Computer Science Tutor — *Oct 2023 - Present*

- Led exam reviews and tutoring sessions in DSA, OOP, and systems programming (***Java***, ***C***, ***Python)*** for 120+ Computer Science students

Projects

AI Portfolio Website, LLM Application — *May 2025 - Jul 2025*

- Built an AI-powered personal website using ***Next.js***, featuring a ChatGPT-style chatbot powered by ***OpenAI's GPT-4o completions API*** to answer recruiter questions, alongside static pages for career overview, resume, and contact form
- Generated vector embeddings from structured portfolio data (bio, skills, projects, etc.) using ***OpenAI's text-embedding-ada-002 API***, and implemented a schema-aware RAG pipeline with cosine similarity for semantic retrieval of context chunks
- Implemented dynamic top-k chunk retrieval based on query intent and similarity thresholds, with fallback logic to handle out-of-scope queries to ensure robust GPT-4o completions
- Tuned the system prompt to define assistant behavior, tone, Markdown formatting, and factual accuracy constraints; leveraged completions API's temperature parameter to reduce hallucinations and trimmed chat history to optimize token usage
- Developed a fully responsive chat interface using ***React.js*** and ***CSS Modules*** with precise layout control and real-time UI state updates
- Implemented streaming UI via a typewriter effect with support for graceful interruptions using AbortController and fallback messaging on cancellation, reducing user-perceived latency
- Enhanced user experience with session-based message persistence, light/dark mode support, and resolution of mobile-specific rendering issues via conditional media queries and physical device testing

NittanyBusiness, Full-Stack Development — *Jan 2025 - Apr 2025*

- Designed and developed a B2B e-commerce procurement platform targeting SMEs with inefficient workflows and limited supplier access
- Built frontend and backend using ***Next.js***, implementing key features such as user authentication, product listings, and order management
- Developed and optimized RESTful APIs for real-time access to product and transaction data, improving system responsiveness and enabling seamless user interactions
- Designed a fully normalized ***MySQL*** database schema to support relational queries, reduce redundancy, and improve performance
- Wrote complex ***SQL*** queries to retrieve and aggregate order, inventory, and user data for backend processing and admin dashboards

Peer-to-Peer File Sharing System, Distributed Systems — *Sep 2024 - Nov 2024*

- Designed a P2P file-sharing network with a central server and multiple peers in ***Python***, enabling efficient, decentralized file distribution
- Developed a chunk-based system to split large files and allow simultaneous downloads from multiple peers, significantly improving download speed, with multi-threading to handle parallel connections
- Integrated file integrity checks using hashing functions to ensure the accuracy of downloaded chunks and discard corrupted data

Dynamic Storage Allocator, Memory Management — *Jan 2024 - Mar 2024*

- Developed a dynamic memory allocator for ***C programs*** with custom implementations of malloc, realloc, and free functions, optimizing space utilization and throughput through block header/footer optimization, free block coalescing, and size-based segregated linked lists
- Created a heap consistency checker to track and resolve heap inconsistencies in debugging mode, ensuring reliable memory management

Technical Skills

AI/LLM Tools: OpenAI API, Cursor, Prompt Engineering
Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, Swift, Verilog
Technologies: React.js, Next.js, Node.js, Express.js, MongoDB, Git, HTML, CSS, SwiftUI, ARKit, RealityKit, I2C